

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
6	(a)	(i)		Lay out grid/ lay out two tapes (1) Random number {table/generator}/ use (10 sided) dice} to give coordinates (1) Avoids bias/ better representation of the whole area/ avoid user generated influence (1)	3			3		3
		(ii)		Secondary succession (1) colonised prior to being cleared, so {seeds/nutrients/ humus} already present(1) (secondary succession) changes environment and allows other species to grow (1)	2	1		3		
		(iii)		Increased (1) More habitats/ food sources/ niches (1)		2		2		
	(b)	(i)		Carry out nitrogen fixation (1) Nitrogen gas converted to nitrogen containing compounds/ ammonium ions(1) NOT nitrate Used by plants to make amino acids/proteins/DNA (1)	2	1		3		
		(ii)		A. (Leghaemoglobin) {combines/ binds} with the oxygen present (1) NOT leghaemoglobin absorbs oxygen B. Leghaemoglobin gives {anaerobic/ oxygen free/ low oxygen} conditions (in the root nodule)(1) C. <i>Rhizobium</i> only fixes nitrogen in {anaerobic/ oxygen free/ low oxygen } conditions(1)			3			